PRAHARI

a digital twin per bed, for the cardiac step-down ward

ARCHITECTURE

1 · WHAT THE WARD HAS

STATIC · EHR

Admission record

Age, sex, body surface area
Diagnosis, aetiology, LVEF, NYHA
BNP, creatinine, Na, K, Hb, HbA1c
Home medication
CYP2D6 genotype → drug response

sets priors and patient constants

DYNAMIC · BEDSIDE MONITOR

Charted every 4 hours

HR · BP · SpO_2 · resp. rate (monitor)

JVP · oedema · weight (ward round)
→ identify congestion (wet / dry)

Capillary refill time
→ identifies perfusion (warm / cold)

Noisy, rounded, ~10% of rounds missed

no networked telemetry, no catheter

AT HOME · WEARABLES

A ₹2,000 band + a scale

Resting HR · HRV · sleep · steps · weight
read from an Apple Health export

same heart → days to decompensation

2 · ASSIMILATE

Learned inverse

Gradient boosting trained on
6,000 simulated physiologies —
no patient data needed

→ fast first guess

Joint trajectory fit

Every reading fitted at once,
weighted by its measurement noise,
older readings weighted less

bounded least squares + priors

Out

Contractility, resistance, volume,
lung water, arterial compliance

+ trends: is the heart failing?

+ confidence: can it say
"I don't know"?

3 · THE TWIN

PHYSICS · NOT A BLACK BOX

Lumped-parameter heart

$SV = Ees(EDV - V_0)/(Ees + Ea)$

Ventricular–arterial coupling
Saturating Frank–Starling curve
Congestion penalty on contractility
Baroreflex on heart rate and SVR
Cardiorenal fluid balance
Pulmonary filtration → SpO_2

A runnable copy of one patient

① state fitted to this patient
② simulates forward in time
③ forks under any decision
④ re-syncs at every reading, and
keeps running between them

reads out as a Stevenson profile:
wet/dry × warm/cold

4 · FORK AND DECIDE

COUNTERFACTUAL ENGINE

Every drug is a parameter lever

Diuretic → volume · nitrate → SVR, preload
Inotrope → contractility · pressor → SVR
Fluids → volume · β-blocker → HR (×CYP2D6)

Onset, plateau and washout per agent

Act now · act in N hours · wait

forked from the patient as they are now

DECISION LAYER

Behind the early-warning score

NEWS2 already says who is sick.
The twin says what kind of sick —
warm-wet, cold-wet, cold-dry — and
what acting or waiting will do.

Deadline to cold-wet · escalation rule
Ranks the ward: who to see first

advises; never issues orders

CLINICIAN STATION

Ward view — every bed, live traces, NEWS2 beside the twin's reading, ranked

Patient view · haemodynamic map · wait-or-act · trends with forecast band · record · handover brief · the twin at home

FastAPI + hand-written browser client · runs on any laptop · no GPU, no cloud, no patient data needed to run

HOW WE KNOW — SIMULATED GROUND TRUTH, THEN REAL DATA

Simulated ward
Registry-shaped patients,
true physiology known

→

Ward degrader
4-hourly, noisy, rounded,
10% of rounds missed

→

Twin sees
only the chart —
never the truth

→

Scored against the truth
① hidden state at 4-hourly vs hourly charting
② every drug given to twin and to "real" patient
③ round-by-round replay and alarm curves vs NEWS2

Then on real data: 32 heart-failure ICU stays from MIMIC-IV (CVP, cardiac output and PA pressure withheld as answer keys) · 11 Fitbit volunteers read through Apple Health exports